

Syllabus for Endsem. Exam:

1.Getting Started and Formalizing Networks: In this module, introduced to the concept of networks. Define networks and identify how data is transformed and analyzed in a network. formalize networks.
2.Social Network Analysis: In this module, able to discuss the structure of networks and be able to explain how a person can be the center of one. You will be able to discover the different types of language that networks use and be able to identify the three types of network measurements.
3.Analyzing a Network with Software: In this module, begin with a social network analysis lab activity. You will be able to do data wrangling of databases and visualize a network. You will be able to analyze a social network and also be able to examine other social network analysis through case studies.
4.Network Evolution: In this module, identify the different types of social networks. You will be able to discuss what mechanisms generates these different types of networks and you will be able to explain how networks move from being static to dynamic.
5.Growing Networks and Making Predictions: In this module, examine theoretical predictions of networks. You will be able to calculate basic math problems and be able to discuss how to make networks more efficient and stable.
6.Link Prediction Feature based Link Prediction, Bayesian Probabilistic Models, Probabilistic Relational Models, Linear Algebraic Methods: Network Evolution based Probabilistic Model, Hierarchical Probabilistic Model, Relational Bayesian Network. Relational Markov Network. Event Detection Classification of Text Streams, Event Detection and Tracking: Bag of Words, Temporal, location, ontology based algorithms. Evolution Analysis in Text Streams, Sentiment analysis.. Social Influence Analysis Influence measures, Social Similarity - Measuring Influence, Influencing actions and interactions. Influence maximization.

A social network is a structure made up of individuals or organizations (called nodes) that are connected by one or more types of interdependency, such as friendship, kinship, shared interests, professional relationships, or information exchange. There are two main contexts in which "social network" is used:

➤ **Social Science Context:**

- I. A social network refers to the **relationships and interactions** between people or groups.
- II. This concept is used in **sociology and anthropology** to study how people are connected and how information, influence, or behaviors spread within a community.

➤ **Online/Digital Context:**

- I. A social networking service or platform like Facebook, Twitter, LinkedIn, Instagram, etc., allows users to create profiles, connect with others, and share content.
- II. These platforms are a digital extension of traditional social networks, enabling interaction at a much larger and more global scale.

Social networks are divided into two types:

Vertical Social Network: Vertical social networks are those that gather users around a specific topic or activity. Their specification makes them attractive to those who are fans of a particular subject and encourages, in turn, interaction among them. There are social networks that focus on athletics, dogs, hiking or even crochet.

Horizontal social networks

Horizontal networks include those social networks with heterogeneous themes, where there is room for a little bit of everything, and where the saying “variety is the spice of life” applies. Instagram or Facebook are examples of horizontal social networks, places where users share content of any kind, driven by personal interests or tastes.

Professional social networks

Vertical or horizontal, are those social networks whose purpose is to establish work-focused relationships. Personal promotion based on the exposure of a profile and resume, search for specific personnel of interest, cybernetic networking in which to promote business opportunities and union between different companies or projects.

Entertainment social networks

Unlike professional ones, entertainment social networks are **leisure for leisure’s sake**. They offer a window to disconnection (oxymoron) that offers an unfathomable amount of content.

Traditional Social Networks:

Examples: Facebook, LinkedIn, X (Twitter), Threads, Mastodon.

We talked a lot about traditional social networks, so I won’t go too much more into that type. But, as I talked about, there are some shifting trends in how brands use traditional social networks. For example, we can see on LinkedIn and Facebook that people often share articles and long-form posts. This isn’t to say that short-form posts have gone away; we see short-form posts across all

traditional social networks. We can also see that Facebook has evolved to include more engagement options, such as Facebook Marketplace for shopping, Facebook Live, and Facebook Groups. The platform also serves as a directory for businesses to create a storefront in the form of a Facebook Page to promote their products, book services, message customers, and leave a space for reviews. Facebook Pages and Groups are great for brands to engage with their audience and reach new people. LinkedIn, on the other hand, has been a significant player in online professional networking. These traditional social networks offer opportunities for brands to use ads to target their audience and conduct market research through Insights and analytics tools.

Media Sharing Networks

Examples: Instagram, Pinterest, TikTok.

There are many forms of visual content across media sharing networks, such as infographics, images, and short-form and long-form videos. While this network type includes video and picture media sharing, most are now multimedia, utilizing audio, video, and pictures. On Instagram, there are several main ways brands use the platform: video and image sharing on the brand's profile, then Instagram Stories, IGTV, and Reels. People use these sites for entertainment, finding information, and sometimes even shopping. This creates ample opportunities for your brand to reach its audience. Many products have recently gone viral through TikTok. When people can easily purchase through an app, it gives them a direct way to buy before second-guessing. Brands also use these sites to disseminate how-to for their products, collaborate with influencers, promote discounts, and more. When it comes to Pinterest, you can use numerous marketing strategies to grow your brand on the platform. Visual content is an engaging way to connect with customers and intrigue them to follow and learn more about your brand – and, eventually, buy your products. Like with traditional social networks, ads perform well on these platforms, especially when brands use insights and analytics to back their strategies.

Discussion Networks

Examples: Discord, Reddit, Quora.

When people have specific questions or want to discuss a topic, sometimes platforms like Facebook or Instagram aren't the place to go. There can be many reasons why people view and create content across platforms. But when it comes to sites like Discord or Reddit, users are typically there for answers and to connect with communities of people with similar interests. People also tend to go to these platforms for research. Discussion Networks allow brands to answer questions and connect with their communities. Discord alone offers opportunities to create customer community groups, events, and live streams. Sometimes, hopping into a discussion about your brand or industry can help clarify topics and show you are willing to engage with your audience. Your effort to meet your audience where they like to communicate can go a long way in creating long-term customers.

Social Blogging Networks

Examples: Medium, Tumblr, Blog Meets Brand, WordPress.com.

When a picture or video is not the correct form of content for a particular topic, blogs can be an excellent way to share information with your audience. Sometimes information is better shared through a blog. For example, it may better explain a subject and help readers understand more complex topics. Blogs are beneficial for content from thought-leaders. They can expand upon important information or news. Your brand can also repurpose, expand upon, or share what it creates on its site through social blogging networks. For newer brands, it's a great way to develop your voice and build your audience. These are additional platforms, besides their websites, for brands to share their insights on what matters to them and their target audience.

Review Networks

Examples: Yelp, Glassdoor, Google Business Profile, Trip Advisor.

The most important thing for brands regarding review networks is online reputation management. Quickly responding to both positive and negative reviews – in a polite and empathetic way – is crucial for retaining a positive online reputation. Reading online reviews has become essential to the customer journey, and your brand should not ignore them. Claiming listings across various review networks and creating a plan to manage them can help improve your brand's image. It can also show potential and current customers how your brand responds to critiques. It can also provide insight into areas of improvement. As a result, it's an excellent and sometimes underutilized resource.

How networks move from being static to dynamic?

1. Static Network:

- A static social network is one where the structure (nodes and edges) **does not change over time**.
- It represents a **snapshot** of relationships at a particular moment.
- Example: A friendship graph showing connections between classmates at the beginning of the year.

2. Dynamic Network:

- A dynamic social network is one that **evolves over time**.
- Nodes (individuals) and edges (relationships) can **appear, disappear, or change in strength**.
- Example: A professional network like LinkedIn where people continuously add new contacts, update roles, or unfollow others.

Transition from Static to Dynamic:

1. **Time Factor Introduced:**
 - Dynamic networks are modelled across **time steps** or **continuous time**, allowing changes in structure to be recorded.
2. **Real-World Events Influence Change:**
 - People **make new friends, end relationships, join groups, or change behavior**—these are all dynamics reflected in the network.
3. **Technology and Data Tracking:**
 - With digital platforms, it's easier to **track interactions over time** (e.g., messages sent, likes, comments), enabling dynamic analysis.
4. **Weight and Direction Can Vary:**
 - In dynamic networks, relationships can vary in **intensity (weight)** or **direction** over time. E.g., frequent interaction between two users today may decline next month.
5. **Modeling Tools Evolve:**
 - Researchers use **temporal graphs, dynamic graph theory, and time-stamped data** to study how networks form, grow, and dissolve.

Example: Imagine a WhatsApp group:

- **Static View:** The list of members and their connections at one moment.
- **Dynamic View:** Who joins or leaves the group, how message frequency changes, who becomes central over time—all tracked across days or months.

Evolution analysis in text streams

Evolution analysis in text streams refers to studying how topics, patterns, or events **change over time** in a continuously generated flow of text data or to the process of studying how topics, patterns, or events **emerge, change, and disappear over time** in continuous flows of text data, such as social media posts, news articles, emails, or forum discussions.. Text streams come from sources like:

- Social media posts (Twitter, Facebook)
- News articles
- Customer reviews
- Research papers
- Chat logs

This is a key part of **text mining** and **temporal data analysis**.

Key Concepts:

1. **Text Stream:**
 - A continuous flow of unstructured text data arriving over time (e.g., tweets, news feeds, chat logs).
2. **Topic Detection & Tracking:**
 - Identifying new or evolving **topics** from the stream.
 - Tracking how topics **grow, shift, or fade**.
3. **Temporal Dynamics:**
 - Understanding **when** a topic becomes popular (bursts) and how long it remains relevant.
 - Detecting **trends** and **seasonal patterns**.
4. **Concept Drift:**
 - Words or phrases might **change meaning** over time, requiring adaptive models.
 - E.g., "Corona" might mean a beer brand earlier, but post-2020, it likely refers to COVID-19.
5. **Event Detection:**
 - Identifying **real-world events** from sudden changes in word usage (e.g., natural disasters, elections).
6. **Tools & Techniques:**
 - Uses **NLP, machine learning, clustering, and temporal analysis**.
 - Examples: Dynamic Topic Models (DTM), Online LDA, time-series clustering.

Applications:

- **Social media monitoring**
- **News trend analysis**
- **Customer feedback analysis**
- **Crisis detection**

Sentiment analysis

Sentiment analysis is the classification of texts according to the emotion they convey, such as positive, negative, or neutral. It is a subtask within the broader field of opinion mining and can be used to improve recommendation systems and support collaborative work in social computing. However, it faces challenges in detecting sarcasm and determining the specific people or things liked or disliked in a text.

or

Sentiment analysis is a specific subtask within the broad area of opinion mining; in short, the classification of texts according to the emotion that the text appears to convey. Sentiment analysis typically classifies texts according to positive, negative and neutral classifications; so that “This movie is great!” is classified as positive, while “This movie was too long and I got bored half way through” is classified as negative. Opinion mining and sentiment analysis tools, depending on the implementation, often suffer from a few key problems. In particular:

- It is not always clear which of the people or things referenced within a given text are liked or disliked: “The movie had some decent acting but I can’t forgive the use of Papyrus font for the end credits”.
- Sarcasm is difficult to detect “Great movie – nearly as much fun as watching paint dry”.

Outcomes from a sentiment analysis process may be used to improve recommendation systems; for example, by identifying and reranking contentious products to detect controversy and disagreement. This is potentially a useful ability in automated summarisation tasks, where a range of viewpoints may exist. In the field of social computing, sentiment analysis is envisaged to be useful in supporting collaborative work. Some of the limitations of sentiment analysis arise from the source material. People are more likely to express positive than negative feelings in social environments, and are more likely to express extreme than average feelings.

Sentiment analysis corresponds to the process of identifying the sentiment associated with a piece of text. It usually relies on applying machine learning techniques to classify documents based on a collection of features extracted from the text using other NLP techniques, such as the presence of certain words or the coverage of some topics. In the context of ITS, sentiment analysis can be applied, for example, to large scale collections of tweets in order to determine how people feel about the transport system in general, or certain aspects of it in particular. There are various packages that provide sentiment analysis functionality, such as the “RSentiment” package of R or the “nltk” package of Python. Most of these, actually allow you to train the user to train their own sentiment classifiers, by providing a dataset of texts along with their corresponding sentiments. However, pretrained versions also exist. For example, in Python one can easily use “nltk” package to identify the sentiment in texts.

Social Influence Analysis Influence measures

A social network may be represented as a set of existing social bodies (individuals or organizations at the nodes) and sets of ties/interactions (the connection between nodes) among them. Every society includes several types of networks such as farmer-networks, labour-networks, student-networks, business networks, etc. There are many ways of measuring centrality. The most used concept is the degree of centrality. This measurement counts direct connections. However, this approach is not always effective, as many direct connections may be inactive. Another important centrality measure is betweenness centrality. In this concept, several shortest paths between any pair of nodes through the specified node is considered for measuring betweenness centrality. In this technique, the strength

of the path is not analysed. Other centrality measurements in the literature include closeness centrality, and eigenvector centrality. The k-shell centrality measure attempts to capture the idea of coreness (how close a focal node is to a core group, i.e. to most well-connected nodes) in the network.

The growing use of social networks has attracted many researchers, academics, and organizations to explore social network research topics, including the influence analysis. Influence analysis and its spread on social networks have an important application value by allowing to analyse and explain people's social behaviors. It also provides a theoretical basis for decision making. However, there are still some challenges to work on: there is no mathematical formula of influence; it is difficult to identify the parameters to measure the influence; and, the large amount of data generated by social networks, makes it difficult to analyse and, consequently, to determine the influence. An influence analysis study covers the study of influence properties such as influence evaluation metrics and algorithms, influence maximization, and social data collection and big data analysis.

Influence measures based on walks between pair of users.

Influence measures based on walks between pair of users	Description
Katz centrality	Katz centrality allows not only direct links received by a user but also popularity or status of users sending links to him to be included in his score. Further, the status of each, who has link with these users in turn, should also be used for calculating scores in social network. The equation that defines it is as follows: $C_{Katz} = ((I - \alpha A^T)^{-1} - I)\vec{I}$, Where, I is the identity matrix, $\vec{I}$ is a vector of size n (n is the number of nodes) consisting of ones. A^T denotes the transposed matrix of A and $(I - \alpha A^T)^{-1}$ denotes matrix inversion of the term $(I - \alpha A^T)$. Through Katz measure, most influential node or individual positive tie network can be found who has connections with most of the other users and can influence or affect other users with his decisions or activities. This measure is similar to PageRank algorithm and eigenvector centrality.
Hubbel Measure	Hubbel measure corresponds to the flow of influence through interpersonal links in social networks as input and output channels. The Hubbel measure has structural as well as functional significance. The structural significance of index is in identifying cliques and functional significance is in computation of status. This measure is similar to Katz centrality, the Katz measure uses an identity matrix (each node is connected to itself) while the Hubble measure does not
Bonacich Power Measure	In social networks, the most central user is not always the most powerful one. In order to distinguish between power and centrality, was proposed a set of measures given by $c(\alpha, \beta)$. The parameter β is used to reflect the degree and direction (positive or negative) in which individual user status depends upon status of other users in network. Bonacich power measure is useful in valued and signed graphs, negative ties and positive ties networks.

Influence maximization algorithms

Influence maximization algorithms	Description
Greedy-based algorithms	The study of greedy algorithms is based on hill-climbing greedy algorithm, in which each option can provide the highest value of the impact of the node used to the local optimal solution to approximate the global optimal solution [8]. Some examples of greedy-based algorithms are present bellow: 1) Target wise greedy algorithm based on the potential-based node-selection strategy. This algorithm does not have good results in an initial phase, but it can cover more nodes in a later phase of diffusion; 2) Community-based greedy algorithm was proposed to reduce the cost in terms of execution time. It is based on the IC model; 3) Upper bound-based lazy forward algorithm has been proposed to discover top-k influential nodes. This algorithm sets new limits to significantly reduce the number of Monte Carlo simulations, particularly in the initial phase
Heuristic-based algorithms	According to the computational complexity of the greedy-based algorithms, several heuristic algorithms have been proposed to reduce the solution time and obtain more efficiency of the algorithm. These algorithms select nodes iteratively based on a specific heuristic, instead of computing the marginal gain of the nodes in each iteration. In contrast, its accuracy is relatively low . A proposed algorithm was Two-phase Heuristic Algorithm (TPH). This algorithm is composed of two phases: each node has its offline probability of a given product; therefore, the consideration of local-based maximization cannot focus only on the network topology, but also on the offline property of each node.

Social Similarity - Measuring Influence, Influencing actions and interactions

Social similarity refers to the **degree of resemblance** between individuals or nodes in a social network based on various factors like interests, behaviors, opinions, or connections. A central problem for social influence is to understand the interplay between similarity and social ties. A lot of research has tried to identify influence and correlation in social networks from many different aspects: social similarity and influence; marketing through social influence, influence maximization ; social influence model and practice through conformity, compliance and obedience, and social influence in virtual worlds.

Common measures:

- **Jaccard Similarity:** Based on shared friends or interests.
- **Cosine Similarity:** Based on vectors of interests or behaviors.
- **Structural Similarity:** Based on network structure (e.g., number of mutual connections).

Measuring Influence

Influence is the **ability of one individual to affect others' behaviors or decisions.**

Methods to measure influence:

- **Centrality Measures:**
 - **Degree Centrality:** Number of direct connections.
 - **Betweenness Centrality:** How often a node appears on the shortest paths.
 - **Eigenvector Centrality:** Influence of one's connections.
- **Information Flow Models:**
 - Independent Cascade Model, Linear Threshold Model.
- **Engagement Metrics:**
 - Retweets, likes, shares, comments

Influencing Actions and Interactions

Based on social similarity and influence scores, individuals may:

- **Adopt similar behaviors** (e.g., buying a product seen in a friend's post).
- **Form new links** (e.g., friend suggestions based on mutual interests).
- **React similarly** to events (e.g., political opinions or viral trends).

Example:

If person A and person B have high similarity in interests and A frequently posts about a fitness app, B is more likely to also try or mention that app due to both similarity and A's influence.

Real-World Applications:

- **Targeted advertising**
- **Recommender systems**
- **Viral marketing**
- **Opinion dynamics studies**

Influence maximization

A Social Network (SN) is a kind of network that reflects the social structure of its nodes and their interdependency, such as friendship of people, co-authorship of researchers, and collaboration between different parties. A SN can be treated as a complex network, which is made up of individuals in the society and their relationships among the individuals. The scale of the network is usually very large. This kind of complex social structure plays an important role in dissemination and diffusion of information.

The dramatic development of social networks, has attracted more and more people performing various social activities through these networks, such as online chatting and friendship creation through Skype, content publishing or commenting through micro-Blog or WeChat, profile

browsing through the Internet, messaging through smartphones, and online shopping through e-commerce platform.

The basic understanding of social influence is summarized as follows.

- (1) Social influence is a relationship established between two entities for a specific action. In particular, one entity influences the other entity to perform an action. Usually, the first entity is called the influencer, the second entity is called the influencee.
- (2) Social influence is a function of uncertainty. In particular, if the influencer believes that the influencee will perform the action for sure, the influencer fully “affects” the influencee to perform the action and there is no uncertainty; if the influencer believes that the influencee will not perform the action for sure, the influencer does not “affects” the influencee not to perform the action, then there is no uncertainty either; if the influencer does not have any idea of whether the influencee will perform the action or not, the influencer does not influence on the influencee. In this case, the influencer has the highest uncertainty.
- (3) The level of social influence can be measured by a continuous real number, referred to as the social influence value. Social influence can also be represented with uncertainty (e.g., stronger, strong, weak, weaker, etc.).
- (4) The influencer may have different social influence values on the same influencee for the same action. Social influence is not necessarily symmetric. The fact that A influences B does not necessarily means that B also influences A, where A and B are two entities.

Social Influence is roughly defined as follows: Given two individuals u, v in a social network, u exerts the power on v , that is, u has the effect of change the opinion of v in a direct or indirect way.

The properties of influence

Name	Meaning
Dynamic	Influence refers to the impact is not absolute. It can increase or decrease with new experiences (interactions or observations). It may also decay with time. Compared with the old ones, new experiences are more important since the old ones may become irrelevant with the progress of time.
Propagative	Influence is propagative. For example, Bob influences John, who in turn may influence Alice. However, Bob does not know this, Alice can derive some amount of trust on John based on how much she trusts Bob and how much Bob trusts John. This is not to say, however, that trust is transitive. Because of its propagative nature, information can be passed from one member to another in a social network, creating influence chains. Spreading of influence in social network is the basis of “word of mouth” propagation of information for humans.
Transitive	Generally, influence is transitive. If Alice influences Bob and Bob influences John, this implies that Alice influences John indirectly.
Composable	Propagation of influence along social chains allows a member to form some influence on a member not directly connected to her. However, when several chains influence on a member indirectly, then the influencer needs to compose the influence information. For example, Bob is influenced by Alice through several chains in her network. In this case, Alice needs to compose

	the influence information received from different chains to decide whether Bob influence her.
Measurable	The level of influence can be measured by a continuous real number, referred to as the influence value. Influence also can be represented uncertainty.
Subjective	Influence is subjective. For example, Bob gives an opinion about a movie. If Alice thinks Bob's opinions are always good, she will accept Bob's review. However, John may think differently about Bob's opinions and may not accept the review. The subjective nature of influence leads to personalization of influence computation, where the biases and preferences of the influencer have a direct impact on the computed trust value.
Asymmetric	Influence is typically asymmetric. For example, Bob influences Alice, which does not means that Alice also influences Bob. A member may influence another member more than s/he is influenced back.
Event sensitive	Influence may take a long time to build, but a single high-impact event may destroy it completely. This aspect of influence has received even less attention in computer science.

Framework of social influence analysis:

The framework of social influence analysis is given as:.

- Collection of big data from social networks: It is a very important basis of social influence analysis. With the availability of big data generated by online shopping, advertising, instant messaging, mobile communication, it is much easier and convenient for us to collect raw data from both online and offline sources.
- Big data preprocessing: In order to improve the performance and the convenience of processing, it is necessary to remove the irrelevant information on the social influence analysis. In order to protect the privacy of users, it is also necessary to filter out the sensitive information related to privacy protection from the collected data sets.
- Selection of evaluation metrics: It is very important to extract a set of evaluation metrics to measure social influence of each user. As these evaluation metrics are helpful to quantify social influence of each user, and to find easily the most influential top-k nodes. In the existing work, the common evaluation metrics include centrality of each node, interaction frequency, and so on.
- Measuring social influence: According to the extracted evaluation metrics, evaluation model and computing equations are provided to a specific social network. Thus, social influence of each user can be measured, by integrating the computing equations into the collected real-world data sets.
- Design of influence maximization algorithm: The influence maximization algorithm is designed to find the most influential top-k nodes. Most of the existing algorithms are the improved algorithms based on the greedy algorithm.
- Performance analysis on related algorithm or model: Simulation is made to validate performance (e.g., influential range, computational complexity) of the proposed algorithms based on a specific propagation model. For an excellent algorithm, it has a large influential range and low computational complexity.

